

TEST PAPER INSTRUMENTS

- 1) Which of the following is correct when the variation is west?
- True North is East of Magnetic North.
 - Magnetic North is West of Compass North.
 - True North is West of Magnetic North.
 - Compass North is west of Magnetic North.
- 2) When using a direct reading magnetic compass in the northern hemisphere?
- A longitudinal acceleration on a Westerly heading causes an apparent turn to the South.
 - A longitudinal acceleration on an Easterly heading causes an apparent turn to the North.
 - A longitudinal deceleration on a Westerly heading causes an apparent turn to the North.
 - A longitudinal acceleration on an Easterly heading causes an apparent turn to the South.
- 3) What is the dip angle at the South Magnetic Pole?
- 0
 - 64
 - 90
 - 180
- 4) At the magnetic equator?
- Deviation is zero.
 - The isogonal is an agonic line.
 - Dip is zero.
 - Variation is zero.
- 5) Which of the following is the correct conversion from True to Compass?
- | T | V | M | D | C |
|--------|----|-----|----|-----|
| a) 130 | 2W | 128 | -1 | 127 |
| b) 130 | 2E | 128 | -1 | 129 |
| c) 130 | 2E | 132 | -1 | 133 |
| d) 130 | 2W | 132 | -1 | 133 |
- 6) The horizontal component of the earth's magnetic field?
- Weakens with increasing distance from the magnetic poles.
 - Weakens with increasing distance from the nearer magnetic pole.
 - Is approximately the same at all magnetic latitudes less than 60° .
 - Is approximately the same at magnetic latitudes 50°N and 50°S .
- 7) What is the maximum value of dip that can be caused by the vertical component of terrestrial magnetism?
- 45°
 - 60°
 - 90°
 - 180°
- 8) Which of the following is correct when the variation is West?
- True North is East of Magnetic North.

- b) Magnetic North is West of Compass North.
- c) True North is West of Magnetic North.
- d) Compass North is West of Magnetic North.

9) A line drawn on a chart which joins all points where the value of magnetic variation is zero is called an?

1. Agonic line. 2. Aclinic line. 3. Isogonal. 4. Isotach.

10) How is the direct reading magnetic compass made Aperiodic or dead beat?

- a) Using a pendulous suspension system for the magnetic assembly.
- b) Using the lowest possible viscosity liquid in the compass.
- c) Using short magnets.
- d) Positioning the magnetic assembly mass close to the compass point and using damping wires.

11) Isogonals converge at the?

- a) Magnetic equator.
- b) North magnetic pole only.
- c) North and South magnetic poles only.
- d) North and South geographic and magnetic poles.

12) Why does the value of magnetic variation on a chart change with time?

- a) Movements of the magnetic poles, cause it to increase.
- b) Movements of the magnetic poles, cause it to increase or decrease.
- c) Increases in the magnetic field, cause it to increase.
- d) Reductions in the magnetic field cause it to decrease.

13) What is the compass heading if true track is 348° , drift is 17° left, variation is 32° W, and deviation is 4° E?

- a) 009° .
- b) 033° .
- c) 339° .
- d) 337° .

14) The value of magnetic variation?

- a) Must be 0° at the magnetic equator.
- b) Varies between a maximum of 45° East and 45° West.
- c) Cannot exceed 90° .
- d) Has a maximum of 180° .

15) At what point on the earth is a magnetic compass most effective?

- a) AT the geographic equator.
- b) Close to the magnetic North Pole.
- c) Close to the magnetic south pole.
- d) Approximately midway between the magnetic poles.

16) Which of the following is true of the value of magnetic variation?

- a) It is 0° at the magnetic equator.
- b) It varies between 45° East and 45° west.

- c) It cannot exceed 90° .
- d) It cannot exceed 180° .

17) What is the compass heading if true track is 358° drift is 17° left, variation is 27° East, and deviation is 19° W?

- a. 009° . b. 001° . c. 359° . d. 347° .

18) An Air Data Computer (ADC)?

- a) Measures position error in the static system and transmits this information to ATC to provide correct altitude reporting.
- b) Transforms air data measurements into electric impulses driving servo motors in instruments.
- c) Is an auxiliary system that provides altitude information in the event that the static source is blocked.
- d) Converts air data measurements given by ATC from the ground in order to provide correct altitude and speed information.

19) In An Air Data Computer (ADC), aeroplane altitude is calculated from?

- a) The difference between absolute and dynamic pressure at the fuselage.
- b) Measurement of outside air temperature (OAT).
- c) Measurement of elapsed time for a radio signal transmitted to the ground surface and back.
- d) Measurement of absolute barometric pressure from a static source on the fuselage.

20) In a non-pressurized aircraft, if one or several static pressure ports are damaged, there is an ultimate emergency means for restoring a practically correct static pressure intake?

- a) Breaking the rate-of climb indicator glass window.
- b) Slightly opening a window to restore the ambient pressure in the cabin.
- c) Descending as much as possible in order to fly at a pressure as close to 1013.25 hPa as possible.
- d) Calculating the ambient static pressure, allowing for the altitude and QNH and adjusting the instruments

21) The QNH is by definition the value of the?

- a) Altimeter setting so that the needles indicate zero when the aircraft is on ground at the location for which it is provided.
- b) Atmospheric pressure at the level of the ground over flown by the aircraft.
- c) Altimeter setting so that the needles of the altimeter indicate the altitude of the location for which it is given.
- d) Atmospheric pressure at the sea level of the location for which it is given.

22) With a pitot probe blocked due to ice buildup, the aircraft airspeed indicator will indicate in descent a?

- a) Increasing speed.
- b) Fluctuating speed
- c) Decreasing speed.
- d) Constant speed.

23) After an aircraft has passed through a volcanic cloud which has blocked the total pressure probe inlet of the airspeed indicator, the pilot begins a stabilized descent and finds that the indicated airspeed?

- a) Increases steadily.
- b) Decreases abruptly towards zero.
- c) Decreases steadily
- d) Increases abruptly towards VNE.

24) The static pressure error of the static vent on which the altimeter is connected varies substantially with the?

- a) Static temperature.
- b) Mach number of the aircraft.
- c) Deformation of the aneroid capsule.
- d) Aircraft altitude.

25) The pressure altitude is the altitude corresponding?

- a) In standard atmosphere, to the reference pressure P_s .
- b) In ambient atmosphere, to the pressure P_s prevailing at this point.
- c) In standard atmosphere, to the pressure P_s prevailing at this point.
- d) In ambient atmosphere, to the reference pressure P_s .

26) The response time of a vertical speed detector may be increased by adding a?

- a) Correction based on an accelerometer sensor.
- b) Bi-metallic strip.
- c) Return spring.
- d) Second calibrated port.

27) The density altitude is?

- a) The altitude of the standard atmosphere on which the density is equal to the actual density of the atmosphere.
- b) The temperature altitude corrected for the difference between the real temperature and standard temperature.
- c) The pressure altitude corrected for the relative density prevailing at this point.
- d) The pressure altitude corrected for the density of air at this point.

28) The hysteresis error of an altimeter varies substantially with the?

- a) Time passed at a given altitude.
- b) Mach number of the aircraft
- c) Aircraft altitude.
- d) Static temperature.

29) When flying from a sector of warm air into one of colder air, the altimeter will?

- a) Under read.
- b) Be just as correct as before.
- c) Show the actual height above ground.
- d) Over read.

30) When accelerations on an easterly heading in the Northern hemisphere, the compass card of a direct reading magnetic compass will turn?

- a. Anti – clockwise giving an apparent turn toward the south.
- b. Clockwise giving an apparent turn toward the north.
- c. Clockwise giving an apparent turn toward the north.
- d. Anti – clockwise giving an apparent turn toward the north.

31) At sea level, on a typical servo altimeter, the tolerance in feet from indicated must not exceed?

- a) +/-60 feet.
- b) +/-75 feet.
- c) +/-30 feet.
- d) +/-70 feet.

32) The altitude indicated on board an aircraft flying in an atmosphere where all the atmosphere layers below the aircraft are cold is?

- a) Equal to the standard altitude.
- b) Lower than the real altitude.
- c) The same as the real altitude.
- d) Higher than the real altitude.

33) The purpose of the vibrating device of an altimeter is to?

- a) Allow damping of the measurement in the unit.
- b) Reduce the hysteresis effect.
- c) Reduce the effect of friction in the linkages.
- d) Inform the crew of a failure of the instrument

34) From what is true altitude derived?

- a) Pressure altitude.
- b) Density altitude.
- c) Temperature altitude.
- d) Difference between total pressure and static pressure.

35) What is QNH?

- a) Ambient pressure at the airfield.
- b) Sea level pressure based on ambient pressure at the airfield.
- c) Sea level pressure.
- d) Sea level pressure in the ISA.

36) What is the true altitude of an aircraft if its altimeter indicated 16000 ft when the ambient temperature was -30°C ?

- a) 15200 ft.
- b) 15400 ft.
- c) 16200 ft.
- d) 16400 ft.

37) What will happen to the indicated altitude if an aircraft in level flight passes into a warmer air mass?

- a) Over indicate.
- b) Under indicate.
- c) Not change.

- d) Remain constant only if above the tropopause.

38) What is density altitude?

- a) Pressure altitude corrected for ambient temperature.
- b) True altitude.
- c) Pressure altitude corrected for density changes.
- d) True altitude corrected for density changes.

39) Why are vibrators sometimes fitted in altimeters?

- a) Overcome friction.
- b) Overcome inertia.
- c) Overcome hysteresis.
- d) Reduce lag.

40) If the pitot pipe becomes partly blocked?

- a) The VSI indication will be too low when climbing.
- b) The VSI will be too low when descending.
- c) The VSI will not be affected.
- d) The VSI will be too low when descending and too high when climbing.

41) If the static pipe becomes partly blocked?

- a) The VSI indication will be too high when descending.
- b) The VSI indication will be too high when accelerating.
- c) The VSI indication will be too low when climbing or descending.
- d) The VSI indication will be unaffected.

42) If the choke in the VSI becomes partly blocked?

- a) The VSI indication will be too high when climbing.
- b) The VSI indication will be too low when descending.
- c) The VSI indication will be too high at all times.
- d) The VSI indication will be too high when climbing or descending.

43) If the static pipe becomes partly blocked?

- a) The VSI indication will be too high with descending.
- b) The VSI indication will be too high when accelerating at constant descending.
- c) The VSI indication will be too low at all times.
- d) The VSI indication will be too low when descending.

44) The response rate of a VSI can be improved by fitting a?

- a) Accelerometer system.
- b) Choke system.
- c) Bi-metallic compensator.
- d) Return spring.

45) What does the —barber's pole used on some ASI's indicate?

- a) MMO.
- b) VNE.
- c) TAS.
- d) Temperature and VMO.

46) What speed is V_{NO} ?

- a. That which may only be exceeded with caution and in still air.
- b. That which may never be exceeded
- c. That which may be exceeded only in emergencies
- d. The maximum at which fully control deflection is possible without overstressing the aircraft structure

47) From what is V_{MO} calculated?

- a. TAS.
- b. EAS.
- c. CAS.
- d. RAS.

48) What do the upper and lower limits of the limits of the yellow arc on an ASI represent?

- a. V_{NE} and V_{NO} .
- b. V_{NO} and V_{NE} .
- c. V_{MO} and V_{NE} .
- d. V_{NO} and V_{MO} .

49) If the pitot source and drain become blocked by ice when in cruise flight, how will the ASI respond when descending?

- a. It will under read.
- b. It will over read.
- c. It will read zero in all condition.
- d. It will remain fixed at the reading at which it became blocked.

50) At Msl in the ISA?

- a. $CAS = TAS$
- b. $IAS = TASSS$
- c. $IAS = EAS$
- d. $CAS = TAS$

51) In an ASI system, what does the pitot probe measure?

- a. Total pressure.
- b. Dynamic pressure.
- c. Static pressure.
- d. Ambient pressure.

52) V_{FE} is the?

- a. Maximum speed at which the aircraft is permitted to fly with its flaps extended.
- b. Maximum speed at which the flaps can be extended or retracted.
- c. The minimum speed for flaps up flight.
- d. The maximum speed for flaps up flight.

53) What will be the effect on the ASI the pitot tube of an unpressurised aircraft is fractured and the pitot drain is blocked.

- a. It will over read.
- b. It will under read.
- c. It will give a constant reading.
- d. It will read zero at will speeds.

54) At any given weight or altitude, an aircraft will always lift – off at the same?

- a. CAS.
- b. TAS.
- c. Ground speed.
- d. EAS.

55) CAS is?

- a. EAS corrected for position error and compressibility error.
- b. IAS corrected for position error and instrument error.
- c. TAS corrected for instrument error and ram effect.
- d. IAS corrected for density error and position error.

56) When descending from FL400 and attending to maintain maximum groundspeed, airspeed will be limited by?

- a. V_{NE} then V_{MO} .
- b. V_{NO} then V_{NE} .
- c. M_{MO} then V_{MO} .
- d. V_{MO} then M_{MO} .

57) The calibrated airspeed (CAS) is obtained by applying to the indicated airspeed (IAS)?

- a) A compressibility and density correction.
- b) An instrument and position/pressure error correction.
- c) An antenna and compressibility correction.
- d) An instrument and density correction.

58) During a straight and uniform climb, the pilot maintains a constant calibrated airspeed (CAS)?

- a) The Mach number increases and the true airspeed (TAS) increases.
- b) The Mach number increases and the true airspeed (TAS) is constant.
- c) The Mach number is constant and the true airspeed (TAS) is constant.
- d) The Mach number is constant and the true airspeed (TAS) decreases.

59) For a constant Calibrated Airspeed (CAS) and a level flight, a fall in ambient temperature will result in a?

- a) Lower True Airspeed (TAS) due to a decrease in air density.
- b) Lower True Airspeed (TAS) due to an increase in air density.
- c) Higher True Airspeed (TAS) due to a decrease in air density.
- d) Higher True Airspeed (TAS) due to an increase in air density.

60) When a climbing at a constant Mach number below the tropopause, in ISA conditions, the Calibrated Airspeed (CAS) will?

- a) Decrease.
- b) Increase at a linear rate.

- c) Remain constant.
- d) Increase at an exponential rate.

61) How will mach meter indication respond if an aircraft is flying at constant CAS at FL 270 when it experiences a reduction in OAT?

- a) No change.
- b) Increase.
- c) Decrease.
- d) Increase or decrease depending on TAT.

62) What is the LSS at 30000 ft if ambient temperature is -40° C?

- a) 579 Kts.
- b) 660 Kts.
- c) 584 Kts.
- d) 594 Kts.

63) A mach meter comprises of?

- a) A combination of ASI and altimeter.
- b) A combination of VSI and altimeter.
- c) An ASI with its scale marked in mach numbers.
- d) An altimeter with its scale marked in mach numbers.

64) what is the LSS at msl ISA?

- a) 600 Kts.
- b) 550 Kts.
- c) 750 Kts.
- d) 661 Kts.

65) How will mach meter indication respond if an aircraft passes through a cold front when flying at constant CAS and altitude?

- a) increase
- b) Decrease.
- c) Remain constant.
- d) Increase or decrease depending. On altitude.

66) How will mach meter respond in a constant mach number climb if the static source becomes blocked?

- a) Increase.
- b) Decrease
- c) Remain constant.
- d) Increase or decrease depending on airspeed.

67) The basis properties of a gyroscope are?

1. The gyro's weight.
2. The rigidity in space.
3. The inertia.
4. The high RPM.
5. The precession

The combination of correct statements is?

- a. 3, 4.
- b. 2, 5.
- c. 2, 3, 5.
- d. 1, 3, 5.

68) The characteristics of the directional gyro (DG) used in a gyro stabilized compass system are?

- a. One degree of freedom, whose vertical axis, aligned with the real vertical to the location is maintained in this direction by an automatic erecting system
- b) Two degrees of freedom, whose horizontal axis corresponding to the reference direction is maintained in the horizontal plane by an automatic erecting system.
- c) Two degrees of freedom, whose axis aligned with the vertical to the location is maintained in this direction by an erecting system.
- d) One degree of freedom, whose horizontal axis is maintained in the horizontal plane by an automatic erecting system.

69) When an aircraft has turned 270 degrees with a constant attitude and bank, the pilot observes the following on a classic artificial horizon?

- a) Too much nose-up and bank too high.
- b) Too much nose-up and bank too low.
- c) Attitude and bank correct.
- d) Too much nose-up and bank correct.

70) A turn indicator is built around a gyroscope with?

- a) 1 degree of freedom.
- b) 3 degree of freedom.
- c) 2 degree of freedom.
- d) 0 degree of freedom.

71) At a low bank angle, the measurement of rate-of-turn actually consists in measuring the?

- a) Angular velocity of the aircraft.
- b) Yaw rate of the aircraft.
- c) Pitch rate of the aircraft.
- d) Roll rate of the aircraft.

72) When, in flight, the needle and ball indicator is on the left and the ball on the right, the aircraft is?

- a) Turning left with too much bank
- b) Turning right with not enough bank
- c) Turning right with too much bank
- d) Turning left with not enough bank

73) How many degrees of freedom and what is the spin axis of an attitude indicator?

- a) Local earth vertical two degrees of freedom.
- b) Aircraft lateral axis two degrees of freedom.
- c) Aircraft horizontal axis one degree of freedom.
- d) Aircraft longitudinal axis three degrees of freedom.

74) How will a basic AI respond if an aircraft performs a 270 degree turn at constant bank angle and ROT?

- a) Nose up and bank
- b) Nose down and bank
- c) Nose level and bank
- d) Correct bank and pitch.

75) Aircraft attitude is indicated on?

- A) EICAS/ECAM primary display.
- B) EFIS ND.
- C) EFIS PFD .
- D) All of the above.

76) When turning through 90° at constant AOB and pitch attitude, what will a classic artificial horizon indicate?

- A) Too much nose up and too little bank angle.
- B) B) Too much nose up and too much bank angle.
- C) Too little nose up and too little bank angle.
- D) D) Too little nose up and too much bank angle.

77) When turning through 270° at constant AOB and pitch attitude, what will a classic artificial horizon indicate?

- a) Too much nose up and too little bank angle.
- b) Too much nose up and too much bank angle.
- c) Too little nose up and too little bank angle.
- d) Too little nose up and too much bank angle.

78) The gravity sensing unit in an artificial horizon is used to?

- a) prevent tilting of the gyro
- b) Prevent precession of the gyro.
- c) Erect the gyro.
- d) Provide signals to the autopilot.

79) An artificial horizon has Degrees of freedom in the axis?

- A) Two vertical.
- B) B) Two horizontal.
- C) One vertical.
- D) D) One horizontal.

80) The ball in a serviceable slip indicator isbyandindicate/s the state of slip?

- a) Held central Gravity Does not always.
- b) Positioned Acceleration Does not always.
- c) Held central Gravity Always.
- d) Positioned Acceleration Always.

81) The turn indicator is affected by?

1. AOB. 2. Airspeed. 3. Weight. 4. Altitude.

- a. 1, 2.
- b. 2, 3.
- c. 3, 4.
- d. 4, 5.

82) ROT indications are?

- a. Proportional to TAS.
- b. Proportional to CAS.
- c. Proportional to mass.
- d. Proportional to EAS.

83) When both the needle and ball of a turn and slip indicator are displaced to the right the aircraft is?

- A) Turning right with insufficient bank.
- B) Turning right with too much bank.
- C) Turning left with too much bank.
- D) Turning left with insufficient bank.

84) A rate 1 turn at 120 kts requires?

- A) 10° AOB.
- B) 20° AOB.
- C) 30° AOB.
- D) 40° AOB.

85) The correct turn and slip indications when turning right on the ground are?

- A) Needle and ball right.
- B) Needle and ball left.
- C) Needle right and ball left.
- D) Needle left and ball right.

86) ROT indications depend on?

Airspeed. 2) Mass. 3) AOB.

- A) 1, 2.
- B) 2, 3.
- C) 1, 3.
- D) 1, 2, 3.

87) For a rate one turn at 250 Kts the AOB must be?

- A) 22°
- B) 33°
- C) 44°
- D) 55°

88) Following a left engine failure the pilot of a multi-engine aircraft uses rudder to arrest the yaw, whilst side slipping down track with the wings held level by the ailerons. What will the turn and slip indicator show in this condition.

- a) Both needle and ball central.
- b) Both needle and ball right.

- c) Both needle and ball left.
- d) Needle left and ball right.

89) In the displayed weather modes, the intensities of the returns in ascending order of intensity are?

- a) Yellow, Green, Blue and red.
- b) Yellow, Green, Red and Magenta.
- b) Green, Yellow, Red and Magenta.
- c) Blue, Green, Yellow and Red.

90) Total Air Temperature is Than by an amount which is proportional to

- a) Higher Static air temperature CAS.
- b) Higher Static Air temperature TAS.
- c) Lower Dynamic air temperature CAS.
- d) Higher Static air temperature LSS.

91) Total air temperature is

- A) SAT plus kinetic heating effect.
- B) SAT plus the heating effect caused by shock waves at high mach numbers.
- C) SAT plus skin friction heating effect.
- D) Dynamic heating effect.

92) The formula for TAT is?

- A) $TAT = SAT (1 + (0.2 \times K \times M^2))$.
- B) $TAT = SAT / (1 + 0.2 \times K \times M^2)$
- C) $TAT = SAT / (1 - 0.2 M^2)$.
- D) $TAT = SAT (1 - KM^2)$

93) What types of thermometer are used in modern aircraft?

1. Resistive. 2) Mercury. 3) Capacitive. 4) Inductive. 5) Thermocouple

- a) 1, 2, 4.
- b) 1, 3, 4.
- c) 2, 4, 5.
- d) 1, 2, 5.

94) A thermocouple system employs?

- a) Two dissimilar metals joined at one end and a Wheatstone bridge at the other.
- b) Two dissimilar metal joined at one end and a ratio meter at the other.
- c) Two dissimilar metals joined at both ends and a moving coil instrument.
- d) Two dissimilar metals separated by a dielectric substance at one end and a capacitive gauge at the other.

95) A Direct reading magnetic compass will be affected by?

- a) Soft iron.
- b) Hard iron.

- c) Aluminum.
- d) Soft iron effect hard iron, and hard iron.

96) When landing on a northerly heading a direct reading magnetic compass will indicate?

- a) A westerly turn.
- b) An easterly turn.
- c) No turn.
- d) Rapidly increasing oscillations.

97) Magnetic heading can be calculated from true heading using?

- a) A compass and a map indicating isogon lines.
- b) A compass and calibration card.
- c) A calculator and a deviation card
- d) A compass and a deviation card.

98) An aircraft using a direct reading magnetic compass in the Northern hemisphere starts a Rate 1 turn to the right from 145 C. What will be the indicated heading if it rolls out of the turn after 30 seconds?

- a. It is not possible to answer this question using the information given.
- b. Greater than 235°
- c. Less than 235°
- d. 235°

99) What is the advantage of the remote indicating compass (slaved gyro compass) over the direct reading magnetic compass?

- a. It is lighter.
- b. It is connected to a source of electrical power and so is more accurate.
- c. It senses the earth's magnetic field rather than seeks it, so is more sensitive.
- d. It is not affected by aircraft deviation.

100) The Earth can be considered as being a magnet with the?

- a. Red pole near the north pole of the earth and the direction of the magnetic force pointing straight down to the earth's surface.
- b. Blue pole near the north pole of the earth and the direction of the magnetic force pointing straight up from the earth's surface.
- c. Red pole near the north pole of the earth and the direction of the magnetic force pointing straight up from the earth's surface
- d. Blue pole near the north pole of the earth and the direction of the magnetic force pointing straight down to the earth's surface